World Happiness Report 2025

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Summary of the World Wide Happiness Data of 2025

Below is the summary of the Principal Component Analysis of the Data.

Below is the PCA results and the importance of each component.

```
Importance of components:
                          PC1    PC2     PC3     PC4     PC5     PC6     PC7
Standard deviation     2.4869 1.4441 1.04420 0.97863 0.85178 0.65034 0.54865
Proportion of Variance 0.5154 0.1738 0.09086 0.07981 0.06046 0.03524 0.02508
Cumulative Proportion  0.5154 0.6892 0.78004 0.85985 0.92031 0.95556 0.98064
                           PC8    PC9    PC10      PC11      PC12
Standard deviation     0.44260 0.1865 0.03997 0.0007094 0.0002538
Proportion of Variance 0.01632 0.0029 0.00013 0.0000000 0.0000000
Cumulative Proportion  0.99697 0.9999 1.00000 1.0000000 1.0000000
```

The names of the other 5 data that make up the **`Principal Component Analysis (PCA)`**: which listed as standard deviation, rotation, center, scale, and X as listed below which helps makes the overall principal Component Analysis.

Below is the overall pca results of the data regarding World Wide Happiness as of 2025

```
Standard deviations (1, .., p=12):
 [1] 2.4869341176 1.4440600724 1.0442011014 0.9786282308 0.8517830029
 [6] 0.6503355928 0.5486518620 0.4425988936 0.1865440613 0.0399684858
[11] 0.0007093876 0.0002537566

Rotation (n x k) = (12 x 12):
                                                          PC1         PC2         PC3
Year                                        -0.054955448  0.58391339 -0.02322174
Rank                                         0.389281248  0.10079920 -0.05697153
Ladder.score                                -0.393027870 -0.12298601  0.05177339
upperwhisker                                -0.391912017 -0.12593620  0.05273154
lowerwhisker                                -0.393710073 -0.12000480  0.05079968
Explained.by..Log.GDP.per.capita            -0.311677668  0.27132121  0.09700005
Explained.by..Social.support                -0.306587249  0.21672304  0.07251657
Explained.by..Healthy.life.expectancy       -0.283525129 -0.08939297  0.04203072
Explained.by..Freedom.to.make.life.choices -0.251030132  0.28628695 -0.25021796
Explained.by..Generosity                     0.002223456 -0.28812068 -0.75099488
Explained.by..Perceptions.of.corruption     -0.199647728  0.01786558 -0.53500276
Dystopia...residual                         -0.089224194 -0.55644401  0.24271126
                                                         PC4           PC5
Year                                         0.41071623 -0.0721906164
```

```
Rank                                       -0.05707573  0.0288378163
Ladder.score                                0.08960947 -0.0007800777
upperwhisker                                0.10383386 -0.0030449178
lowerwhisker                                0.07572767  0.0013906386
Explained.by..Log.GDP.per.capita           -0.19495453 -0.1064249076
Explained.by..Social.support               -0.07428486  0.4488494681
Explained.by..Healthy.life.expectancy      -0.57839777  0.1762173713
Explained.by..Freedom.to.make.life.choices  0.39021510  0.0836360755
Explained.by..Generosity                    0.11565460  0.4541049640
Explained.by..Perceptions.of.corruption    -0.21083304 -0.7076215157
Dystopia...residual                         0.46405243 -0.1901733221
                                                    PC6           PC7
Year                                        0.200884617  0.0931393547
Rank                                       -0.112170149 -0.0372479224
Ladder.score                                0.050728782 -0.0028690443
upperwhisker                                0.053162667 -0.0067822114
lowerwhisker                                0.048277194  0.0009435435
Explained.by..Log.GDP.per.capita            0.532124518  0.3791266086
Explained.by..Social.support                0.004207301 -0.7365765350
Explained.by..Healthy.life.expectancy      -0.368311150  0.3375065480
Explained.by..Freedom.to.make.life.choices -0.645711985  0.2506421465
Explained.by..Generosity                    0.314907856  0.1399707319
Explained.by..Perceptions.of.corruption    -0.063139283 -0.3270734719
Dystopia...residual                        -0.020200516 -0.0173071729
                                                    PC8          PC9
Year                                       -0.657632886 -0.007564554
Rank                                       -0.029638930  0.903205856
Ladder.score                               -0.009239230  0.197631623
upperwhisker                               -0.007553904  0.213347649
lowerwhisker                               -0.010859510  0.182348496
Explained.by..Log.GDP.per.capita            0.367457698  0.195016397
Explained.by..Social.support                0.016064806  0.064136608
Explained.by..Healthy.life.expectancy      -0.503055013  0.044624161
Explained.by..Freedom.to.make.life.choices  0.356531712  0.024219570
Explained.by..Generosity                   -0.091291741  0.018486603
Explained.by..Perceptions.of.corruption    -0.068584603  0.036011205
Dystopia...residual                        -0.194627059  0.141329908
                                                   PC10          PC11
Year                                        4.322749e-03 -0.0000222014
Rank                                        1.832398e-02 -0.0001110725
Ladder.score                                1.137605e-02 -0.3116063268
upperwhisker                               -7.066700e-01 -0.3389140834
lowerwhisker                                7.067670e-01 -0.3500894731
Explained.by..Log.GDP.per.capita            3.121061e-03  0.4128131562
Explained.by..Social.support                3.818699e-03  0.3158646971
Explained.by..Healthy.life.expectancy      -1.377688e-02  0.1985478906
Explained.by..Freedom.to.make.life.choices  4.624399e-03  0.1601399817
Explained.by..Generosity                    2.166779e-03  0.0771822993
Explained.by..Perceptions.of.corruption    -1.346207e-05  0.1069684316
Dystopia...residual                         1.933222e-02  0.5593080091
                                                   PC12
Year                                        5.431668e-06
Rank                                       -5.617018e-05
Ladder.score                               -8.250693e-01
```

```
upperwhisker                                3.927873e-01
lowerwhisker                                4.055752e-01
Explained.by..Log.GDP.per.capita            1.117186e-02
Explained.by..Social.support                8.560245e-03
Explained.by..Healthy.life.expectancy       5.363476e-03
Explained.by..Freedom.to.make.life.choices  4.325461e-03
Explained.by..Generosity                    2.100872e-03
Explained.by..Perceptions.of.corruption     2.889700e-03
Dystopia...residual                         1.513889e-02
```

sdev = variance explained by each PC (via std devs).

rotation = loadings (how variables contribute to PCs).

x = the projected data (scores in PC space).

center = what was subtracted from each column so everything is measured relative to its mean.

scale = scaling factors (usually standardization).

Standard Deviation Of The Data

```
 [1] 2.4869341176 1.4440600724 1.0442011014 0.9786282308 0.8517830029
 [6] 0.6503355928 0.5486518620 0.4425988936 0.1865440613 0.0399684858
[11] 0.0007093876 0.0002537566
```

The standard deviation in was how spread out the data is from the mean.

According to the standard deviations of PC 1 (SD = 2.486) has the highest variability compared to PC 12 (SD = 0.00025).

Rotation Of The Data

This is also know as the matrix of variable loadings (also called `eigenvectors`). Each column corresponds to a PC, and the values tell you how strongly each original variable contributes to that PC.

- Positive loading means the variable and PC move in the same direction.
- Negative loading means the variable and PC move in opposite directions.
- Larger absolute values (farther from zero) mean a stronger influence on that PC.

Principle Component 1 `PC 1` has a lot of negative loading which means the variable and PC move in opposite directions.

Principle Component 2 `PC 2` has a balance amount of positive and negative loading.

Principle Component 3 `PC 3` has a lot of positive loading which means the variable and PC move in the same direction.

Principle Component 4 `PC 4` has a lot of positive loading which means the variable and PC move in the same direction.

Principle Component 5 `PC 5` has a balance amount of positive and negative loading.

Principle Component 6 `PC 6` has a lot of positive loading which means the variable and pc move in the same direction.

Principle Component 7 `PC 7` has a balance amount of positive and negative loading.

Principle Component 8 `PC 8` has a lot of negative loading which means the variable and PC move in opposite directions.

Principle Component 9 `PC 9` has a lot of positive loading which means the variable and the PC move in the same direction.

Principle Component 10 `PC 10` has a lot of positive loading and Large amounts of absolute values and has some large absolute variables which is: Year, Lowerwhisker, Upperwhisker, Explained by Log GDP per capital, Explained by Social support, Explained by Freedom to make life choices.

Principle Component 11 `PC 11` has a lot of positive loading which means the variable and the pc move in the same direction.

Principle Component 12 `PC 12` has a lot of positive loading and has the largest amount of absolute values which is listed as the variables contributing to it is: Explained by Healthy life expectancy, Year, Ladder score, Upper whisker, Lower whisker, Explained by Social support, Explained by Freedom to make life choices

```
                                                  PC1         PC2         PC3
Year                                     -0.054955448  0.58391339 -0.02322174
Rank                                      0.389281248  0.10079920 -0.05697153
Ladder.score                             -0.393027870 -0.12298601  0.05177339
upperwhisker                             -0.391912017 -0.12593620  0.05273154
lowerwhisker                             -0.393710073 -0.12000480  0.05079968
Explained.by..Log.GDP.per.capita         -0.311677668  0.27132121  0.09700005
Explained.by..Social.support             -0.306587249  0.21672304  0.07251657
Explained.by..Healthy.life.expectancy    -0.283525129 -0.08939297  0.04203072
Explained.by..Freedom.to.make.life.choices -0.251030132  0.28628695 -0.25021796
Explained.by..Generosity                  0.002223456 -0.28812068 -0.75099488
Explained.by..Perceptions.of.corruption  -0.199647728  0.01786558 -0.53500276
Dystopia...residual                      -0.089224194 -0.55644401  0.24271126
                                                 PC4           PC5
Year                                      0.41071623 -0.0721906164
Rank                                     -0.05707573  0.0288378163
Ladder.score                              0.08960947 -0.0007800777
upperwhisker                              0.10383386 -0.0030449178
lowerwhisker                              0.07572767  0.0013906386
Explained.by..Log.GDP.per.capita         -0.19495453 -0.1064249076
Explained.by..Social.support             -0.07428486  0.4488494681
Explained.by..Healthy.life.expectancy    -0.57839777  0.1762173713
Explained.by..Freedom.to.make.life.choices  0.39021510  0.0836360755
Explained.by..Generosity                  0.11565460  0.4541049640
Explained.by..Perceptions.of.corruption  -0.21083304 -0.7076215157
Dystopia...residual                       0.46405243 -0.1901733221
                                                  PC6           PC7
Year                                      0.200884617  0.0931393547
Rank                                     -0.112170149 -0.0372479224
```

```
Ladder.score                                0.050728782 -0.0028690443
upperwhisker                                0.053162667 -0.0067822114
lowerwhisker                                0.048277194  0.0009435435
Explained.by..Log.GDP.per.capita            0.532124518  0.3791266086
Explained.by..Social.support                0.004207301 -0.7365765350
Explained.by..Healthy.life.expectancy      -0.368311150  0.3375065480
Explained.by..Freedom.to.make.life.choices -0.645711985  0.2506421465
Explained.by..Generosity                    0.314907856  0.1399707319
Explained.by..Perceptions.of.corruption    -0.063139283 -0.3270734719
Dystopia...residual                        -0.020200516 -0.0173071729
                                                    PC8          PC9
Year                                       -0.657632886 -0.007564554
Rank                                       -0.029638930  0.903205856
Ladder.score                               -0.009239230  0.197631623
upperwhisker                               -0.007553904  0.213347649
lowerwhisker                               -0.010859510  0.182348496
Explained.by..Log.GDP.per.capita            0.367457698  0.195016397
Explained.by..Social.support                0.016064806  0.064136608
Explained.by..Healthy.life.expectancy      -0.503055013  0.044624161
Explained.by..Freedom.to.make.life.choices  0.356531712  0.024219570
Explained.by..Generosity                   -0.091291741  0.018486603
Explained.by..Perceptions.of.corruption    -0.068584603  0.036011205
Dystopia...residual                        -0.194627059  0.141329908
                                                   PC10          PC11
Year                                        4.322749e-03 -0.0000222014
Rank                                        1.832398e-02 -0.0001110725
Ladder.score                                1.137605e-02 -0.3116063268
upperwhisker                               -7.066700e-01 -0.3389140834
lowerwhisker                                7.067670e-01 -0.3500894731
Explained.by..Log.GDP.per.capita            3.121061e-03  0.4128131562
Explained.by..Social.support                3.818699e-03  0.3158646971
Explained.by..Healthy.life.expectancy      -1.377688e-02  0.1985478906
Explained.by..Freedom.to.make.life.choices  4.624399e-03  0.1601399817
Explained.by..Generosity                    2.166779e-03  0.0771822993
Explained.by..Perceptions.of.corruption    -1.346207e-05  0.1069684316
Dystopia...residual                         1.933222e-02  0.5593080091
                                                   PC12
Year                                        5.431668e-06
Rank                                       -5.617018e-05
Ladder.score                               -8.250693e-01
upperwhisker                                3.927873e-01
lowerwhisker                                4.055752e-01
Explained.by..Log.GDP.per.capita            1.117186e-02
Explained.by..Social.support                8.560245e-03
Explained.by..Healthy.life.expectancy       5.363476e-03
Explained.by..Freedom.to.make.life.choices  4.325461e-03
Explained.by..Generosity                    2.100872e-03
Explained.by..Perceptions.of.corruption     2.889700e-03
Dystopia...residual                         1.513889e-02
```

Center Of The Data

```
                    Year
            2021.4527650
```

```
                                   Rank
                             73.5725806
                           Ladder.score
                              5.5352477
                           upperwhisker
                              5.6500012
                           lowerwhisker
                              5.4204643
          Explained.by..Log.GDP.per.capita
                              1.2204793
              Explained.by..Social.support
                              1.0768963
     Explained.by..Healthy.life.expectancy
                              0.5428341
Explained.by..Freedom.to.make.life.choices
                              0.5634090
                  Explained.by..Generosity
                              0.1545703
   Explained.by..Perceptions.of.corruption
                              0.1442903
                      Dystopia...residual
                              1.8327776
```

Scale Of The Data

```
                                   Year
                             1.72339334
                                   Rank
                            42.39402184
                           Ladder.score
                             1.12436986
                           upperwhisker
                             1.10720151
                           lowerwhisker
                             1.14266742
          Explained.by..Log.GDP.per.capita
                             0.46417256
              Explained.by..Social.support
                             0.35500444
     Explained.by..Healthy.life.expectancy
                             0.22316704
Explained.by..Freedom.to.make.life.choices
                             0.18000588
                  Explained.by..Generosity
                             0.08678451
   Explained.by..Perceptions.of.corruption
                             0.12026116
                      Dystopia...residual
                             0.62880239
```

The Data Points That Make Up The PCA Of The Data

```
                 PC1          PC2          PC3          PC4           PC5
  [1,] -5.5528041032  1.514192082 -1.137963922  0.324927845 -1.2934976545
```

```
 [2,]  7.3386444294  1.392051515  0.093998621 -1.451030026 -1.3842591402
 [3,]  7.3612743220  0.785656748  0.338035347 -0.975469242 -1.3839265191
 [4,]  6.6209054242  0.025812241  0.861978845 -1.365068472 -0.9935300251
 [5,]  6.9355539652 -1.172100913  0.710849881 -0.493043105 -1.0368526449
 [6,]  6.5799917341 -1.098448336  0.588822629 -1.386714851 -0.1781717830
 [7,] -5.2316134968  1.646036752 -1.497084396  0.086610064 -1.0052606137
 [8,]  0.2198585453  1.227924652  0.196480399  0.110460097  0.4319357212
 [9,]  0.5943629729  0.544055575  0.578782241  0.163305334  0.2813701512
[10,]  0.8015522156 -0.153902233  0.617661575 -0.685223133  0.1829721909
[11,]  1.3984719877 -1.353364366  0.478126550 -0.157638360  0.1313582789
[12,]  1.4560780428 -1.061482946  0.297310277 -1.516104165  0.9234512044
[13,] -4.8614896656  1.369609031 -0.490245218  0.684620935  1.2498265210
[14,]  0.3952880303  0.523353525  0.697216950 -0.754974718 -0.6965379407
[15,]  0.5822546515  0.350386282  1.090157520 -0.512011333 -0.4781901645
[16,]  1.1066810555 -0.409664616  0.886528282 -1.579911813 -0.5083424204
[17,]  2.1040840443 -1.584764342  0.736941822 -1.027068615 -0.4796902196
[18,]  1.4404603631 -1.450120609  0.963416235 -2.223472292  0.1072250698
[19,] -4.9838042149  1.684456499 -1.663525149 -0.163028015 -0.7303938703
[20,] -1.5698919503  1.037768495  0.944375616  0.505633457  0.3414531341
[21,] -1.0323644141  0.650385208  1.039344283  0.449172664  0.1990472544
[22,] -0.9128664110  0.020739570  1.127817001 -0.421354724 -0.0102575002
[23,] -0.4537460320 -1.107383069  1.436135526 -0.010419705 -0.1702279864
[24,] -0.9066829944 -1.085867369  1.455275017 -1.111268048  0.7113180468
[25,] -4.5713760398  1.328636633 -1.063878856  0.116281074 -0.1084023528
[26,] -0.2875633392  1.611695068  0.503515717 -0.263597220 -0.4985101892
[27,]  0.1788124728  1.064283930  0.713788986 -0.189540071 -0.6210920109
[28,]  0.0756215525  0.322493263  0.508129125 -0.905045045 -1.1112713173
[29,]  0.7066984092 -0.758922230  0.423654069 -0.401613579 -1.0591972214
[30,]  1.7585501503 -0.628596378  0.510014328 -1.935364756  0.3656751518
[31,] -3.9031638820  0.979135107  1.033125951  1.693960530  0.2640090310
[32,] -3.7253375271  0.595731478 -1.192451240  0.218537320 -0.2201263613
[33,] -3.4678067801  0.006863505 -1.094800768  0.391471496 -0.1977375308
[34,] -3.6184975266 -0.743858604 -1.333436055 -0.372977335 -0.4596494602
[35,] -3.2377984627 -2.118694824 -1.342864483 -0.010715930 -0.3301068745
[36,] -3.7092208901 -2.133913146 -1.754707300 -1.226307212  0.6228132289
[37,] -4.8502158996  1.959135335 -1.525513368 -0.192974357 -0.4716943777
[38,] -3.3466852377  0.526323891 -0.980090059  0.096898459 -0.3616697774
[39,] -3.3648248104 -0.121292197 -0.531771748  0.365141762 -0.6927996544
[40,] -3.5461188659 -0.743782069 -0.590512328 -0.461303145 -0.8494421224
[41,] -3.3022884234 -2.053907680 -0.477786082  0.110950070 -0.6826519468
[42,] -3.6755734955 -2.222297373 -0.780097190 -0.982108360  0.4238231054
[43,] -4.0509918882  0.989512593  0.229844552  0.363216259  0.5218190944
[44,]  0.8306105326  1.775850401 -0.451205664 -0.207044735 -0.6619500054
[45,]  0.0898706366  1.327203251 -0.084984734 -1.348705660 -1.5943510555
[46,]  0.7445161934 -0.334203396  0.321131384 -0.718895326 -1.5834645629
[47,]  0.4398510563 -0.287697439  0.511346451 -1.887309528 -0.7026501159
[48,] -4.5498873352  2.109072056 -0.944831256 -0.341760900 -0.8068681207
[49,] -1.5970272870  0.865429076 -0.451651881  0.667056640  0.3653637461
[50,] -2.3558953045 -0.504314546 -0.143791616  0.273481475 -0.0089935815
[51,] -1.9456651438 -1.878956980 -0.328593211  0.569774262  0.1633281878
[52,] -1.6105567995 -1.363563696 -0.870695286 -0.973524911  1.4363044066
[53,] -3.1844076892  0.908085907  1.140041525  1.886813415 -0.0488347886
[54,]  2.8037300337  1.843133532 -1.140252206 -0.084465360 -0.9063292512
[55,]  2.4455996549  0.939473130 -0.622958113  0.123175803 -0.8669554978
```

```
 [56,]  0.8868749619 -0.292854217 -0.158506156 -0.049581925 -0.9039958420
 [57,]  1.5813264007 -1.408822222 -0.299758473  0.427968569 -0.7548648827
 [58,]  1.4549481133 -1.033190493 -0.687756939 -0.867634346  0.1041231942
 [59,] -0.6919180055  0.014188371  0.839322587 -1.301061910 -1.4026150131
 [60,]  0.3531336785 -1.046913160  0.963161281 -0.757424122 -0.9901131161
 [61,] -0.0134361702 -1.030165148  0.798813633 -1.894184101 -0.0410399984
 [62,] -3.9844325452  1.618587673 -0.742533598  0.006221383  0.1633667754
 [63,] -3.4234330707  0.849881999 -0.680736163  0.020279165 -0.4588384068
 [64,] -2.9592356891  0.244220839  0.065853742  0.109229154 -0.6430429207
 [65,] -2.6504214617 -0.521729010  1.366773182 -0.666372449 -1.0399155830
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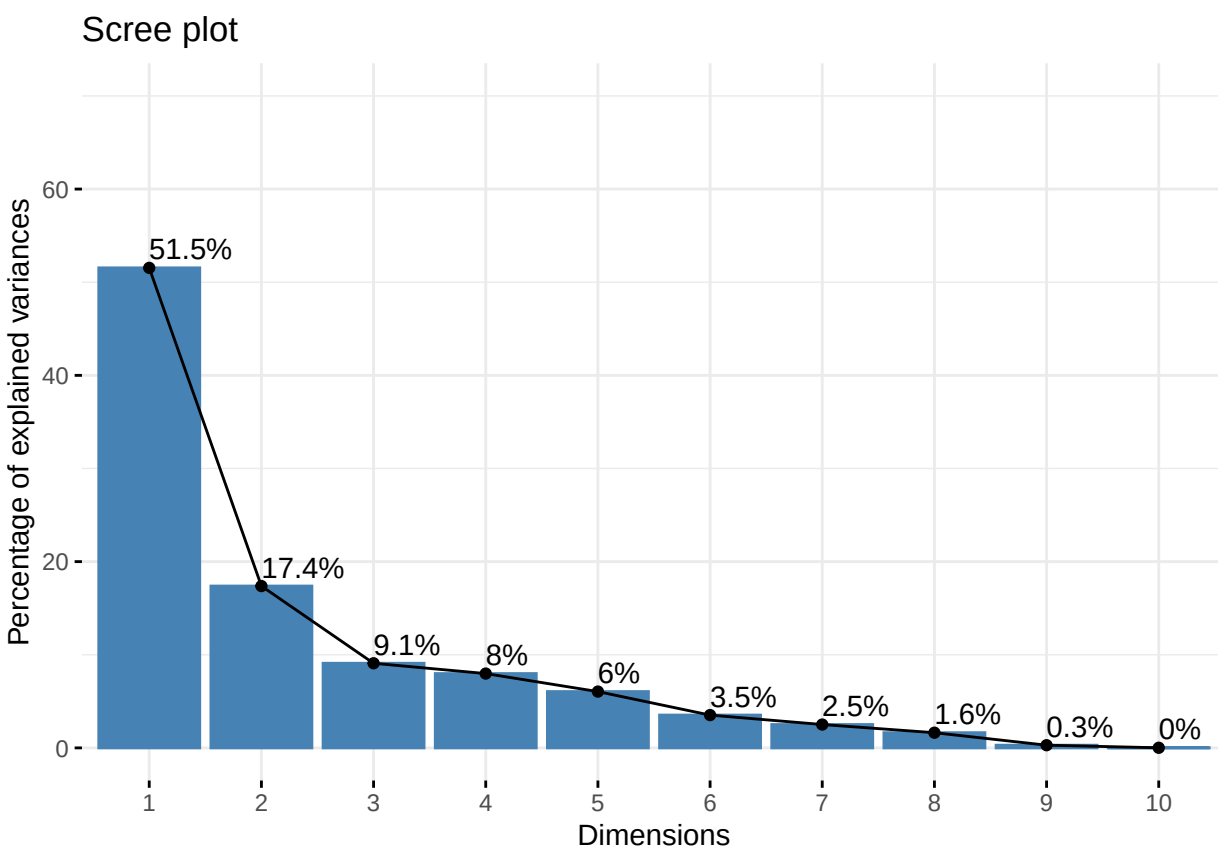
Visualization of the data

Below is a Scree plot of the data

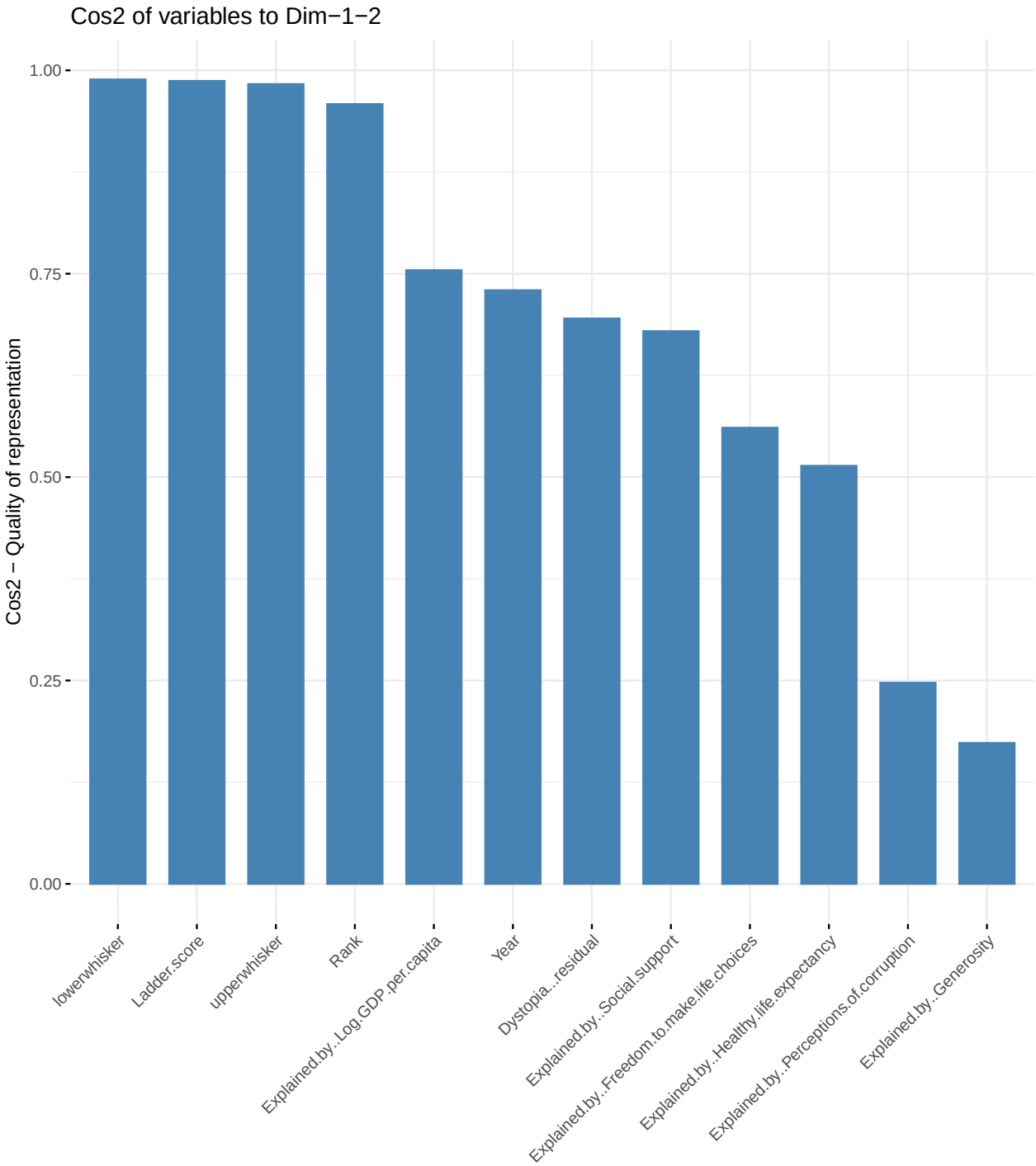


The scree plot shows the total amount of dimension in the data and how each dimension affects the data.

- Dimension 1: Accounts for the most variance in the data at `51.5%`.
- Dimension 2: Accounts for the second highest amount of variance in the data at `17.4`.
- Dimension 3: Accounts for `9.1%` of the variance in the data.
- Dimension 4: Accounts for `8%` of the variance in the data.
- Dimension 5: Accounts for `6%` of the variance in the data.

- Dimension 6: Accounts for `3.5%` of the variance in the data.
- Dimension 7: Accounts for `2.5%` of the variance in the data.
- Dimension 8: Accounts for `1.6%` of the variance in the data.
- Dimension 9: Accounts for `0.3` of the variance in the data.
- Dimension 10: Accounts for the lowest amount of variance in the data which is `0%`.

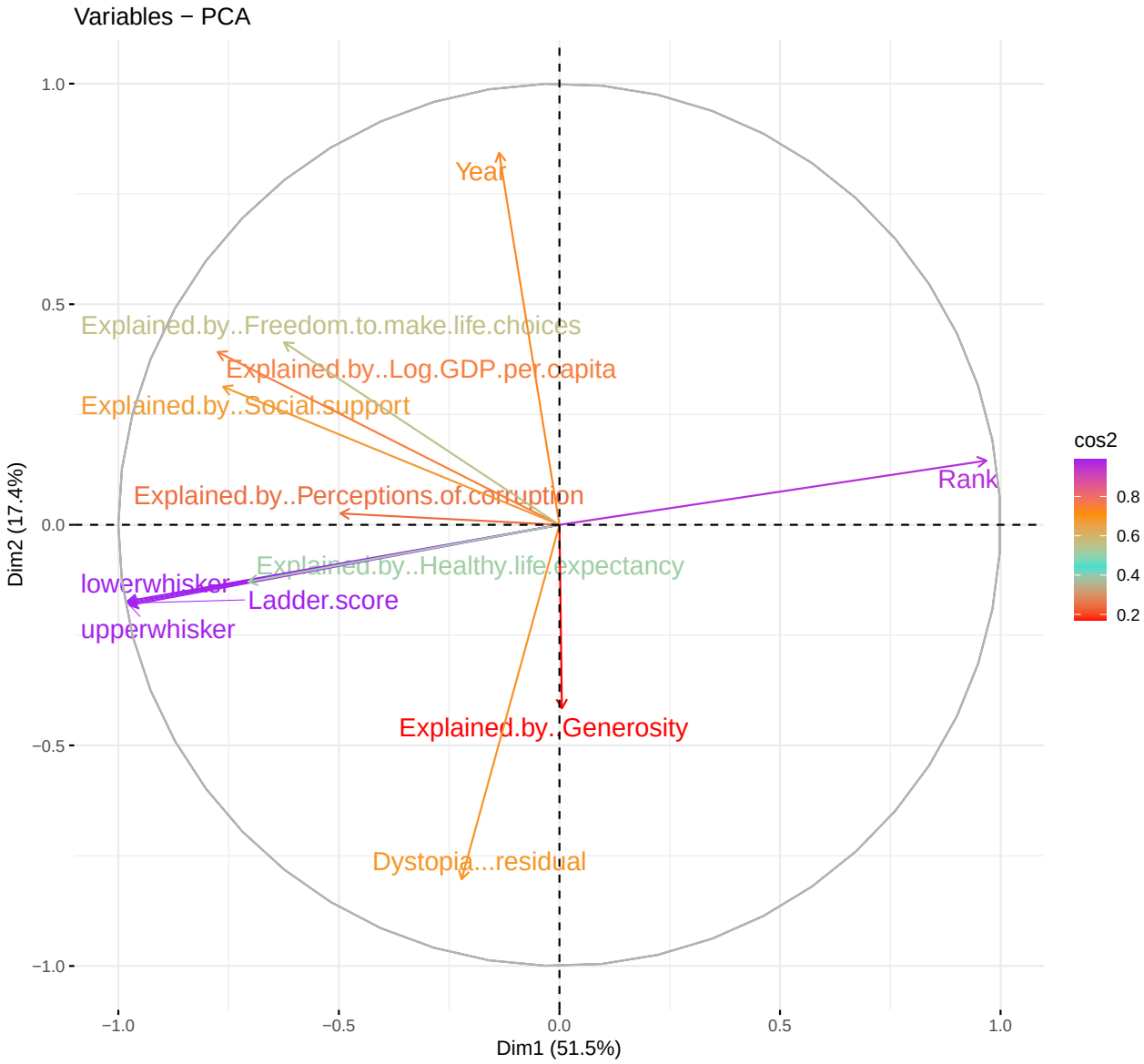
Below is the cosine of the PCA which helps list out which variables contribute to the most variance in the data.



- Lowerwhisker: Contribute to the most variance in the data.
- Ladder Score: Has the second highest contribution to the variance in the data. It is also higher than upperwhisker but lower than lowerwhisker.
- Upperwhisker: Has a higher variance than Rank but it also has a lower variance then ladder.score.
- Rank: Has a higher variance than explain by log gdp per captial but a lower variance than upperwhisker.

- Explained By Log GPD Per Captia: Has a higher variance than Year but a lower variance than rank.
- Year: Has A higher variance than Dystopia residual but a lower variance than explained by log gpg per captia.
- Dystopia Residual: Has a higher variance than explained by social support but has a lower variance than year.
- Explained By Social Support: Has a higher variance than explained by freedom to make life choices but lower than dystopia residual.
- Explained By Freedom To Make Life Choices: Has a higher variance than explained by healthy life expectancy but a lower variance than explained by social support.
- Explained By Healthy Life Expectancy: Has a higher variance than explained by perceptions of corruption but a lower variance than explained by social support.
- Explained By Perceptions Of Corruption: Has a higher variance than explained by generosity but has a lower variance than explained by healthy life expectancy.
- Explained By Generosity: Has the lowest amount of variance in the data frame.

Biplot of the data to show variance

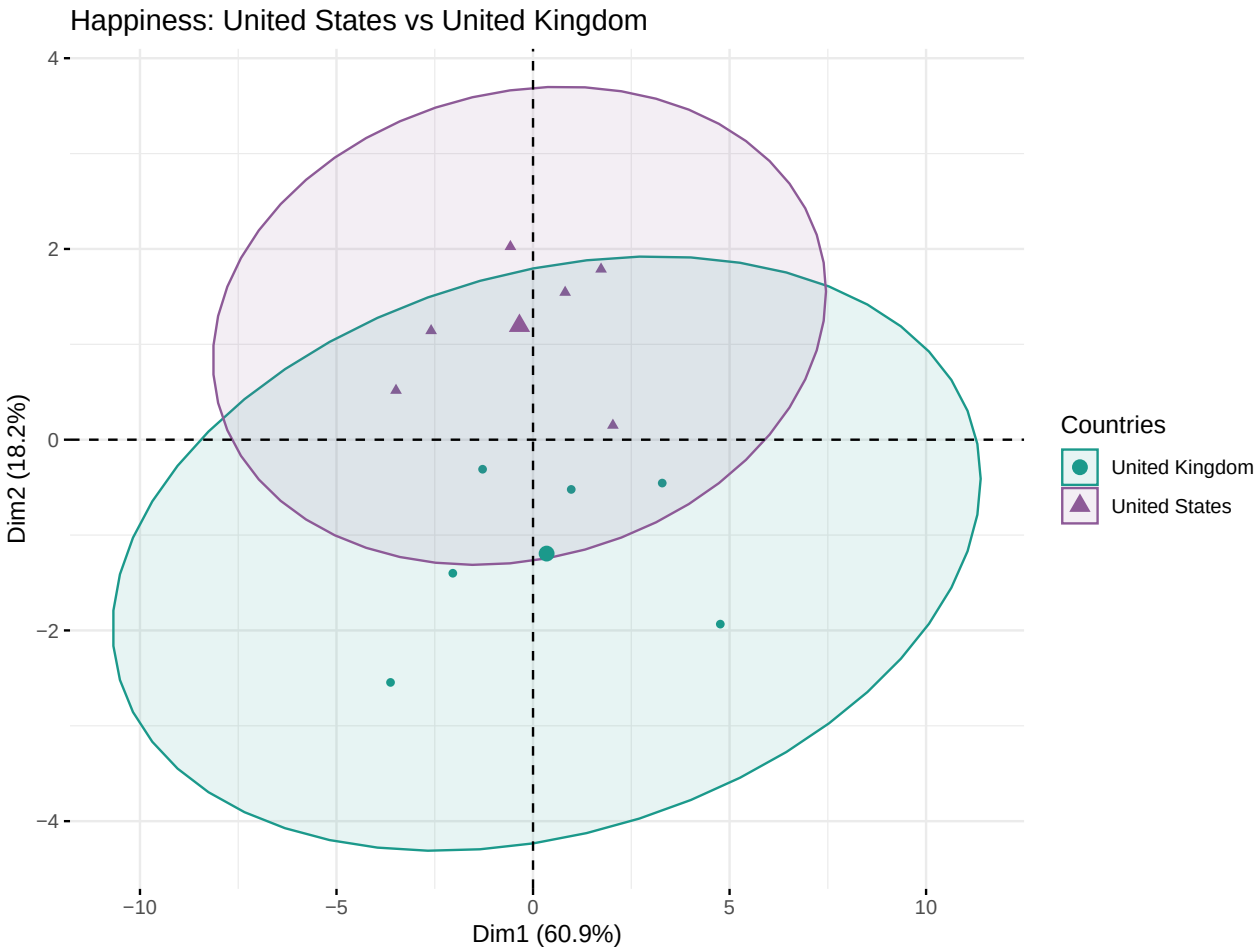


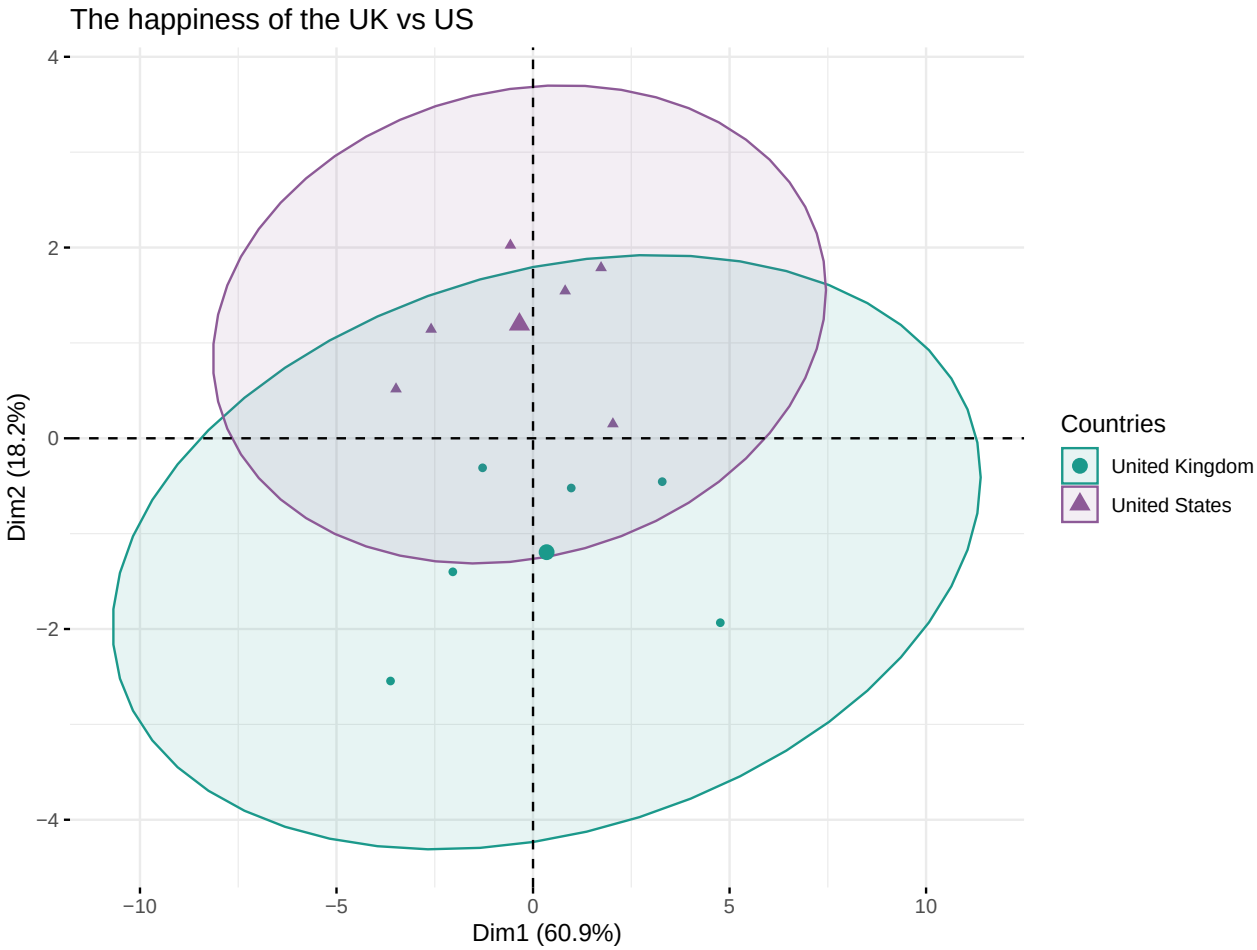
- The variables ladder score, lower whisker, explained by healthy life expectancy, and upper whisker are all 4 highly positively correlated to each other. Also, lower whisker, ladder score, and upper whisker contribute to most of the variance of data along with rank. However, rank is negatively correlated to ladder score, lower whisker and upper whisker since it is going directly opposite direction of ladder score upper whisker lower whisker. These results are shown in purple because the variables which the highest rating.
- Explained by healthy life expectancy has a light greenish greyish color showing that is contributes to the data at a medium amount in terms of variance of the data.
- The variables explained by freedom to make life choices, explained by log gpd per captia, explained by social support, are all highly positively correlated to each other and explained by perceptions of corruption is also positively correlated as well but less correlated. The variables are show cased in orange and dark orange as grey color that their they have an above average correlation to the data compared to the data above.
- Year is negatively correlated to the variables Dystopia residual and explained by generosity and year

and Dystopia residual contributes to a lot of variance of the data less than the variance of ladder score lower whisker, explained by healthy life expectancy. Year and Dystopia have a orange color showing that they have a high correlation to the data

- Explained by generosity has the lowest contribution to the variance in the data that is shown because it has the lowest rating which is red.

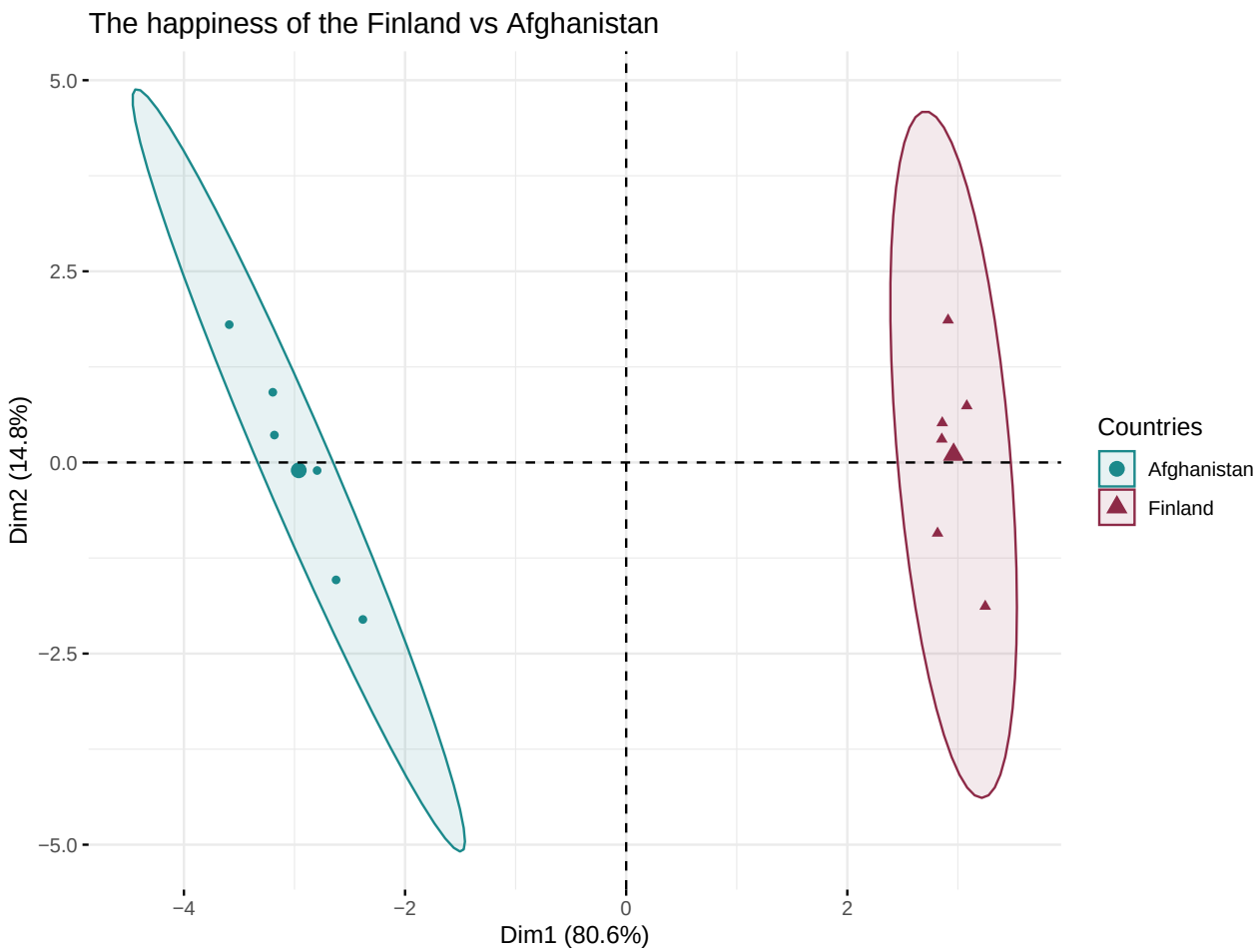
PCA Graphs Results



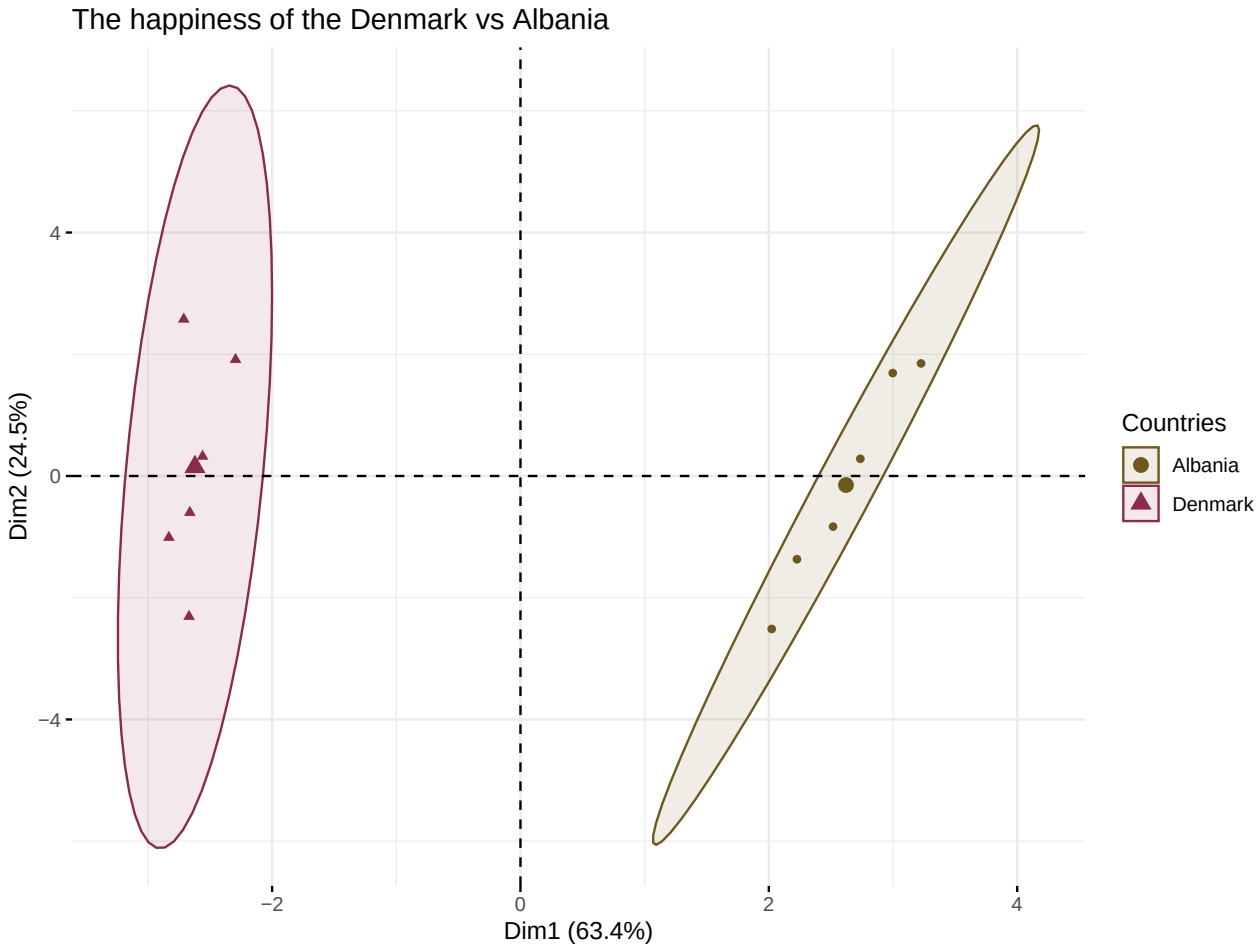


The happines of the United States vs the United Kingdom

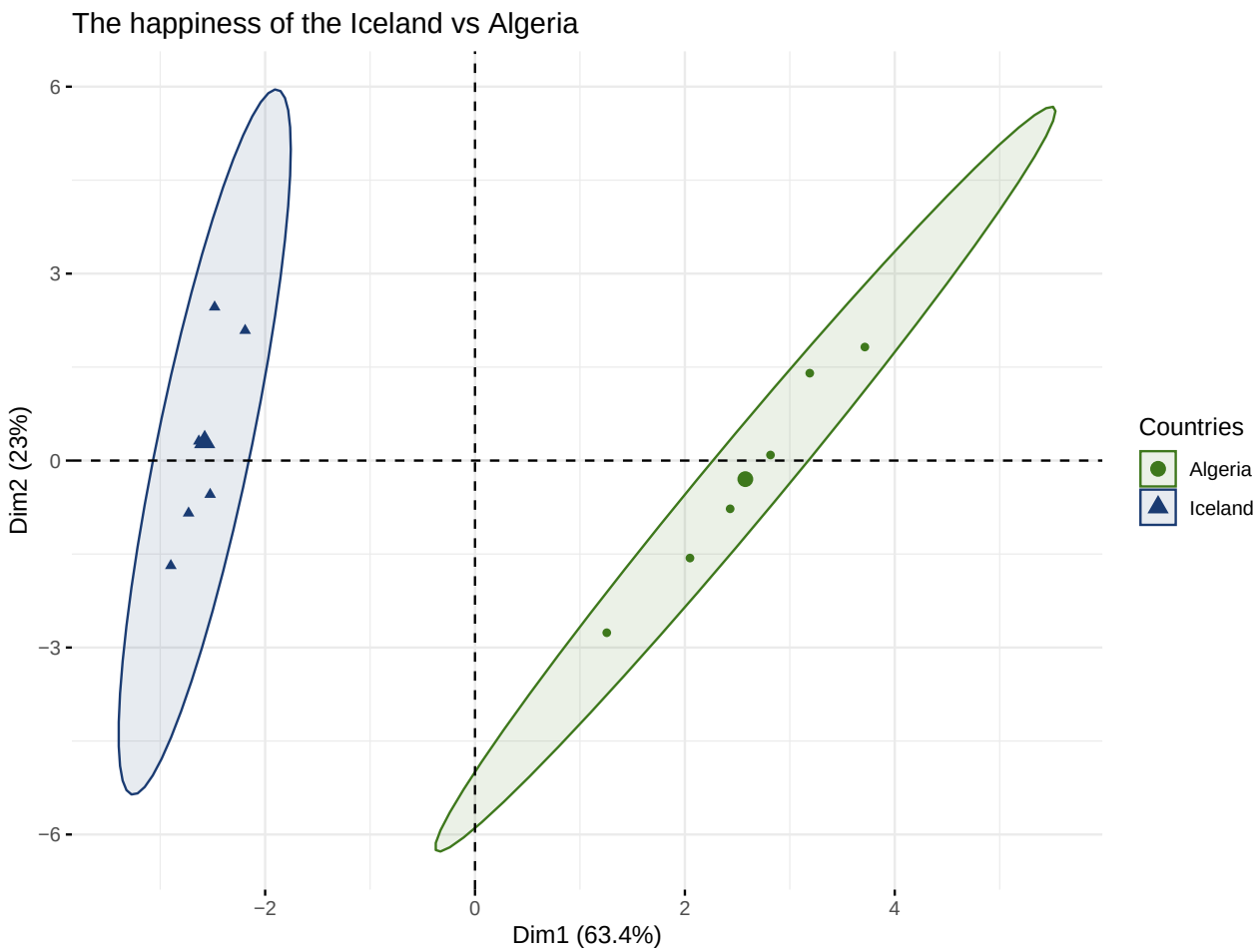
- **United States** exhibits a slightly smaller variance in happiness scores.
- Smaller ellipse suggests consistent happiness levels across the population, indicating that while many Americans report high happiness, there is also a significant portion experiencing lower happiness levels.
- The united states scored higher on the happiness variance because the points ellipse and points are positive for dimension 1(PCA 1) which means the primary happiness value is higher compared to the UK but their is still some people who experience some unhappiness regarding the primary happiness value.
- dimension 2(PCA 2) has a lower mix of positive and negative values so for for the secondary happiness is more mixed for the happiest for the country.
- **United Kingdom** shows considerably greater variance with a Larger ellipse and more spread out data points.
- This indicates that happiness levels among the British population are more spread out and the happiness varies with low amount of happiness and others with a high amount of happiness, with more variation between the highest and lowest reported happiness scores.
- The UK ellipse is also lower on the happiest rating because the cluster is mainly on the negative of dimension 1(PCA 1) which indicates lower primary happiness value.
- Dimension 2 (PCA 2) has a mix in between negative and positive happiness for secondary happiness values.



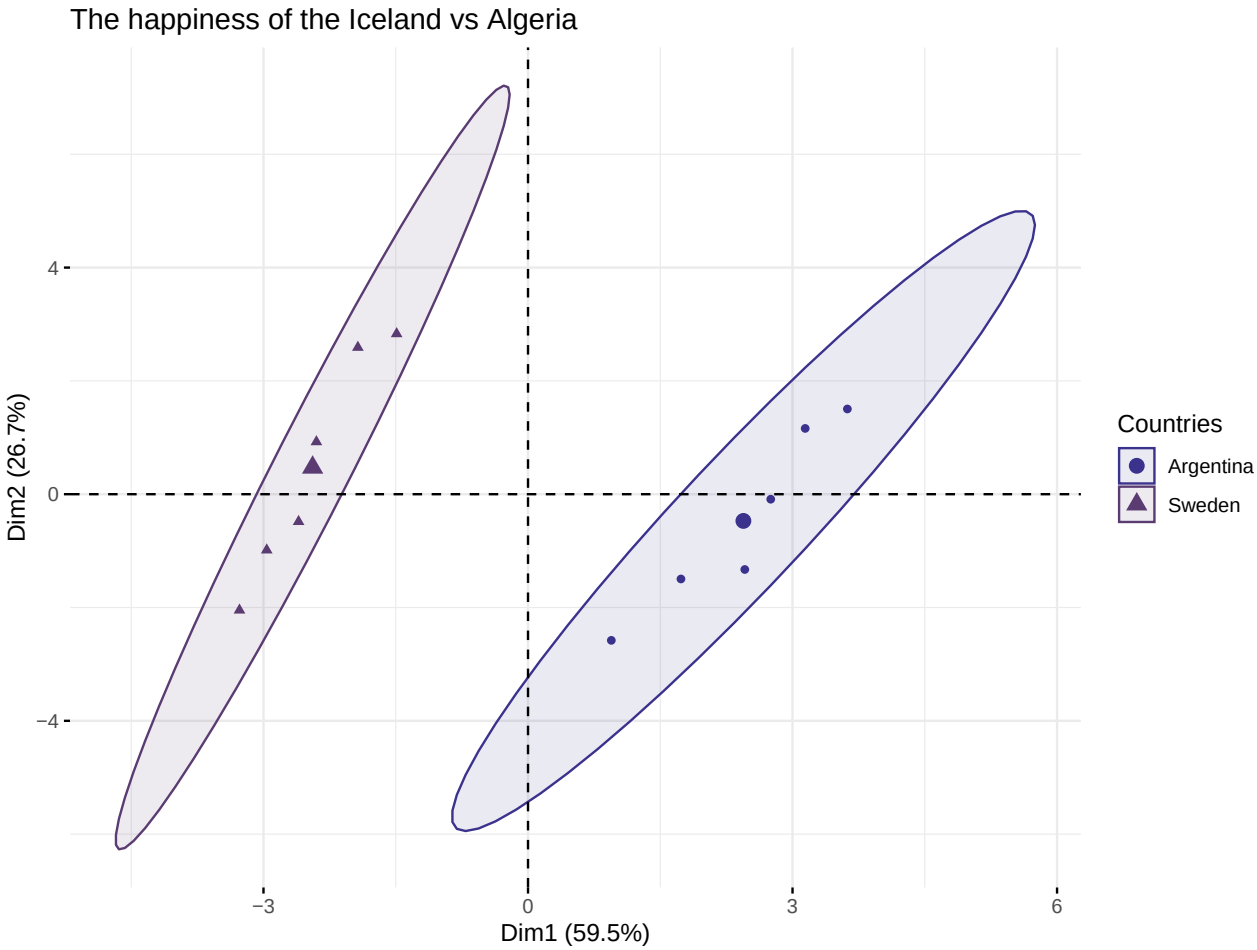
- **Afghanistan** has a more thin wider ellipse and the points are more closely together which symbolizes how this countries happiness is varied at a normal level where some to happy or sad.
- The dimension 1(PCA 1) has a both negative and positive for the primary happiness value.
- The dimension 2(PCA 2) has a negative values only for dimension 2 which means that the secondary value is has a much lower value for happiness
- **Finland** has a little larger ellipse and the points are a little spread out which symbolizes how this countries happiness is much more spread out so some individuals feel more sad while others feel much more happy.
- The dimension (PCA 1) has a both mixed amount of happiness for the primary value of happiness because the happiness is both positive and negative.
- The dimension (PCA 2) is in positive axis which means that the second happiness value contributes a lot to overall happiness.



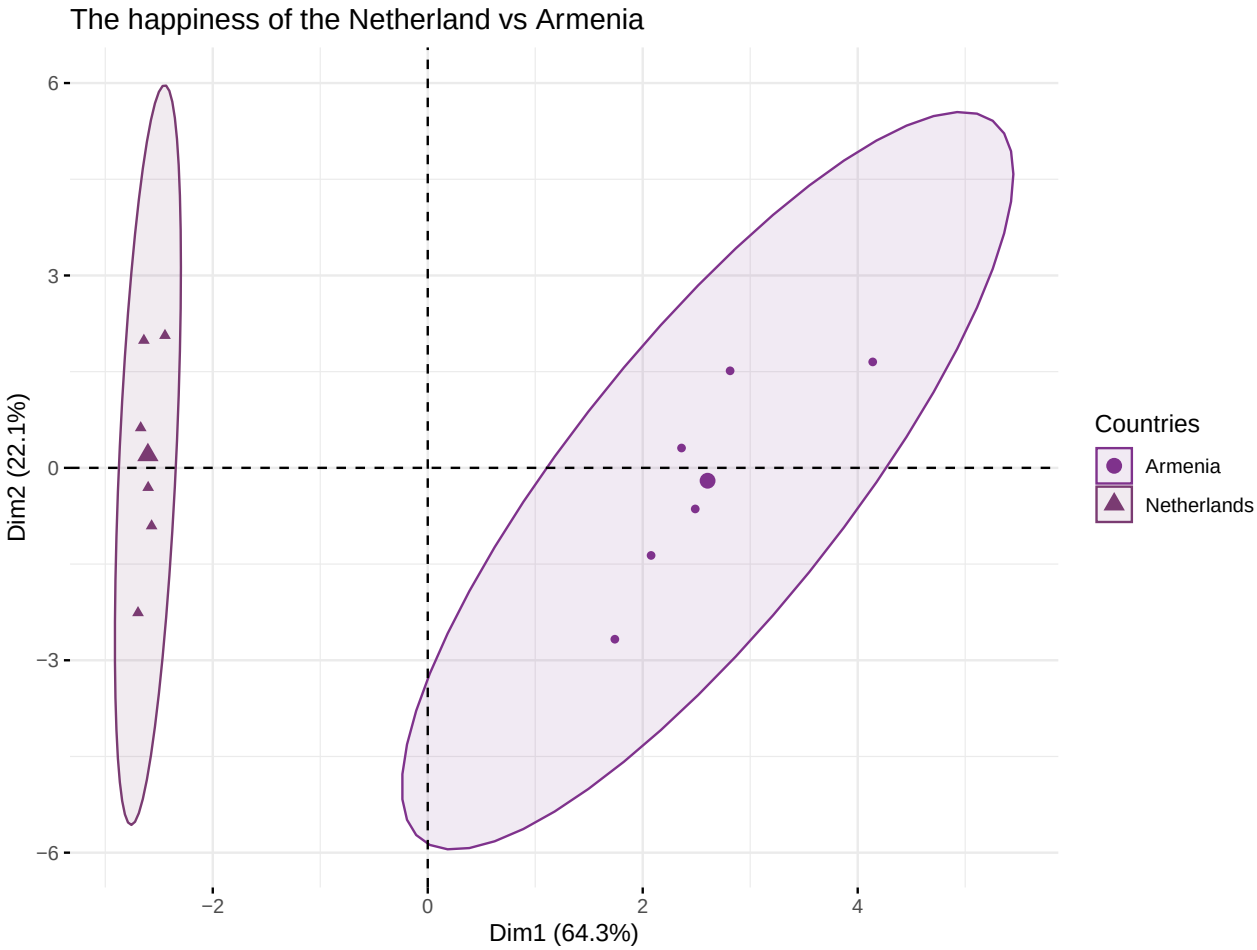
- **Albania** Has an extremely small variance of happiness where the happiness is more consistent and the amount of happiness is that some people are more happy while others show less happiness. The ellipsis is extremely thin that which shows that the variant isn't really varied.
- Dimension 1(PCA 1) has a the mixed amount of happiness because the happiness is both positive and negative for dimension 1 for the primary variable for happiness.
- Dimension 2(PCA 2) has a positive amount of happiness which means that the happiness from the second variable contributes most to the over all happiness for the second variable for happiness.
- **Denmark** Has a much wider amount of variance of happiness where some people are extremely happy while others are much less happy based on the points. The ellipsis have is bigger which demonstrates the bigger varience for the happiness where some are a little more happy while others aren't as happy.
- Dimension 1(PCA 1) has a mixed amount of variance of happiness where some people are extremely happy or others are not every happy base on the primary variable of happiness.
- Dimension 2(PCA 2) has a negative amount of happiness which means that the the secondary variable of happiness contributes the most to the unhappiness based on the second variable for happiness.



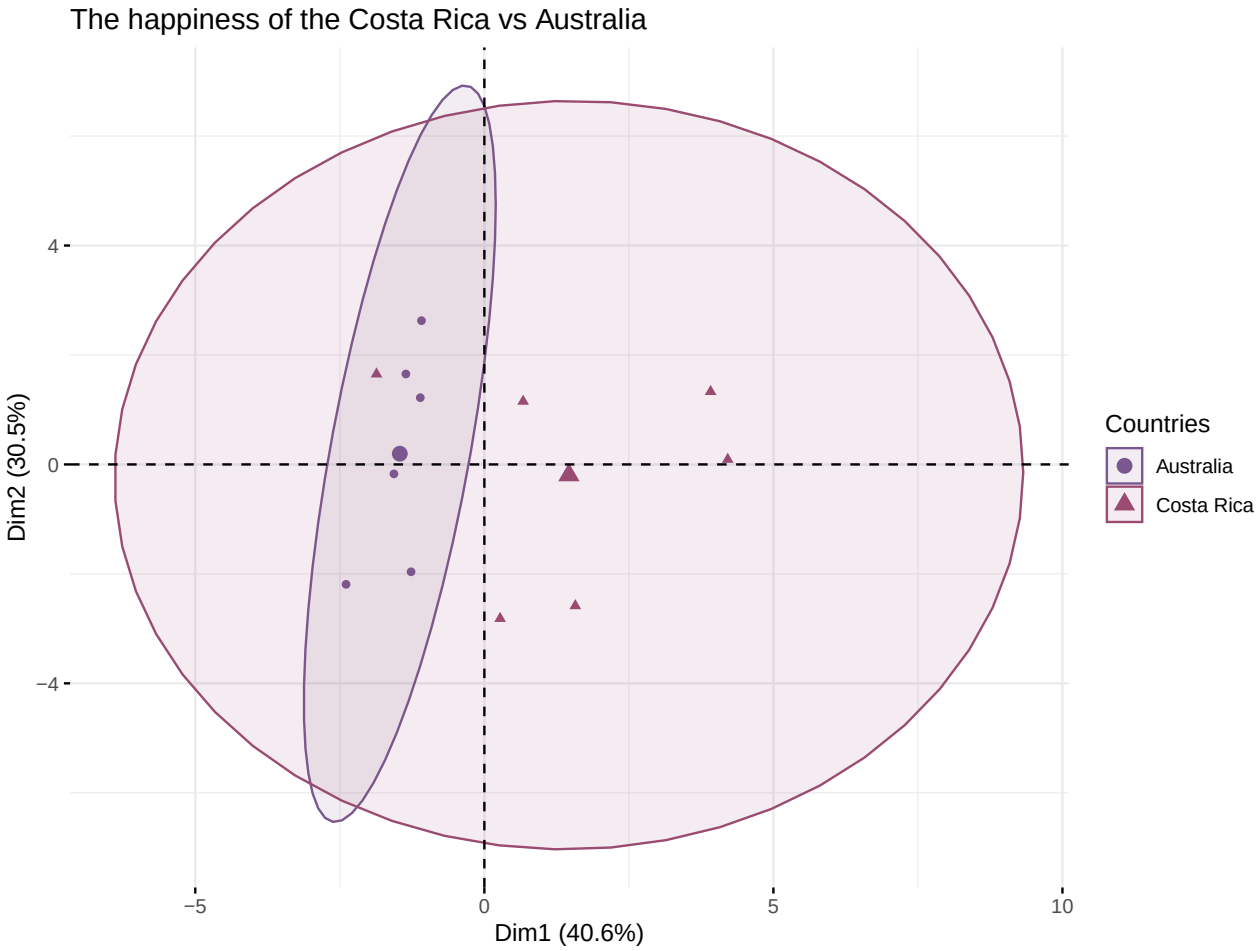
- **Iceland** has a thin ellipse variance where some people experience more happiness while some people are unhappy. This is shown in the points where some are high and others are low.
- Dimension 1(PCA 1) dimension 1 has a more mixed amount of happiness where some is unhappy and others are more happy.
- Dimension 2(PCA 2) dimension 2 has a negative happiness which means that dimension 1 accounts for more happiness over dimension 2's happiness.
- **Algeria** has a extremely thin bigger ellipse variance where of the points have a variance where it happiness rose over time from low to high.
- Dimension 1(PCA 1) dimension 1 has a more mixed balanced amount of happiness because they are negative and positive.
- Dimension 1(PCA 2) dimension 2 has a much higher positive amount of happiness affect compared to dimension 1.



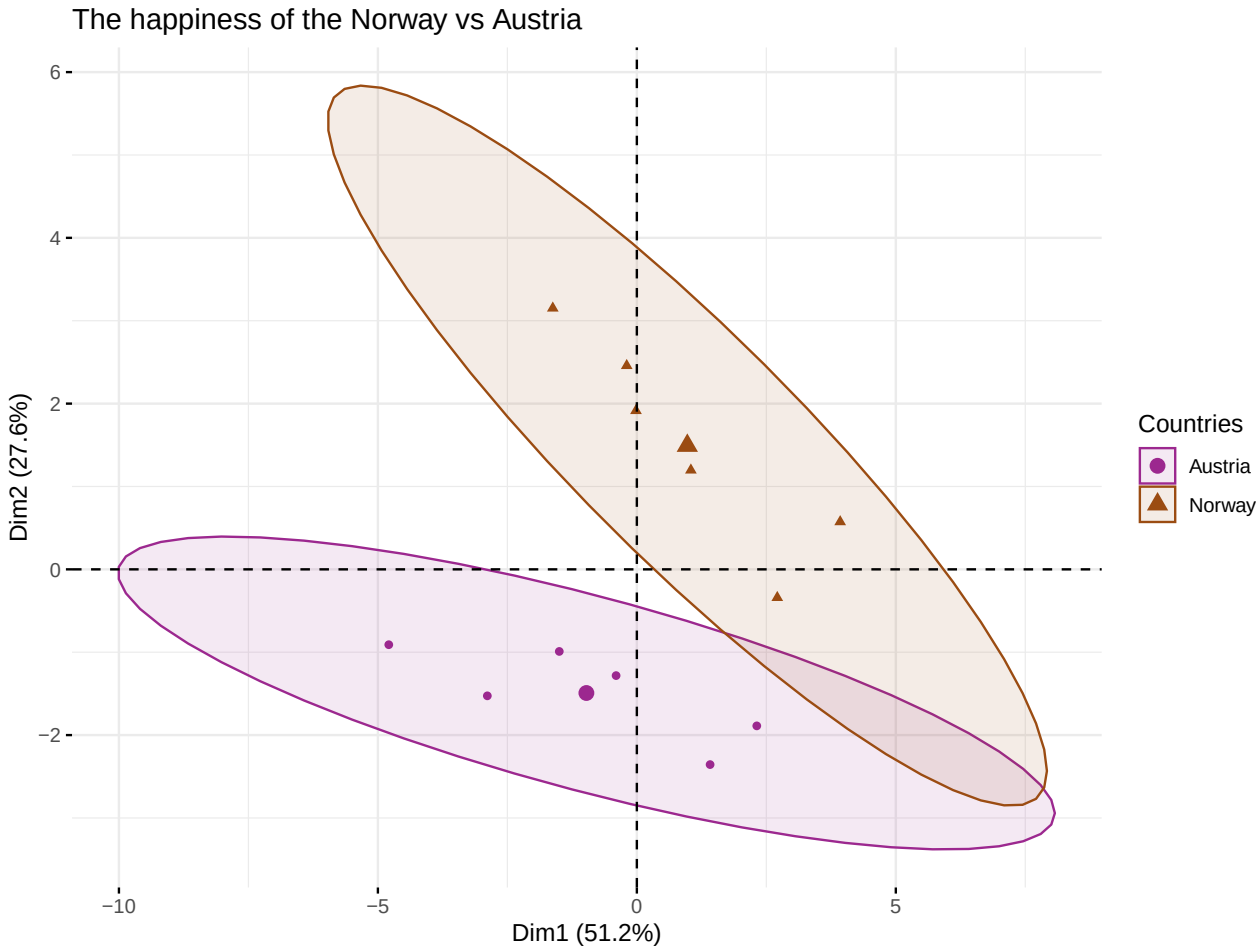
- **Argentina** the ellipse is slightly bigger than the ellipse for Sweden which means that the happiness amount is more varied.
- Dimension 1(PCA 1) has a mixed amount of positive and negative points where most are mixed for happiness amount based on the primary variable for happiness.
- Dimension 2(PCA 2) has a much more positive amount for the happiness for the second most influential factor of happiness.
- **Sweden** the ellipse is more thin than the ellipse for Argentina which means the points will not be as varied.
- Dimension 1(PCA 1) has a more mixed amount of happiness both positive and negative for the primary .
- Dimension 2(PCA 2) has a much higher positive amount of happiness for the secondary happiness variable compared to the primary variable of happiness which was more mixed.



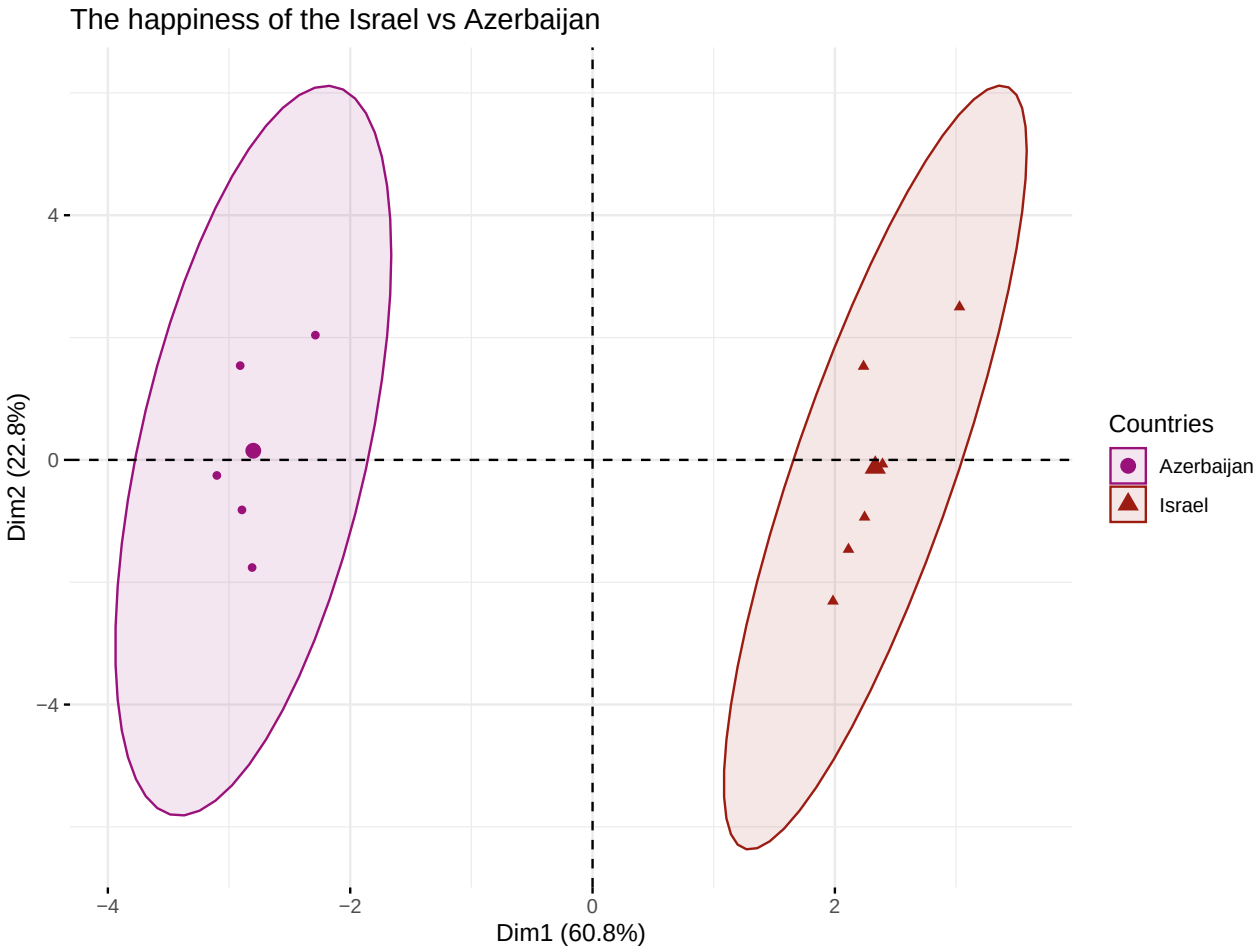
- **Netherlands** has a very thin ellipse which means the variance of the happiness discrepancy will be much smaller.
- Dimension 1 (PCA 1) the variance for happiness is mixed where it is both positive and negative for primary variable for happiness.
- Dimension 2(PCA 2) the variance for happiness is negative no positve variance for dimension 2.
- **Armenia** has a wide big ellipse which means a wider discrepancy.
- Dimension 1(PCA 1) the variance for happiness for the primary variable for happiness is mixed for the average happiness.
- Dimension 2(PCA 2) the variance for the happiness for the secondary variable for happiness is much more positive for the average of the overall happiness.



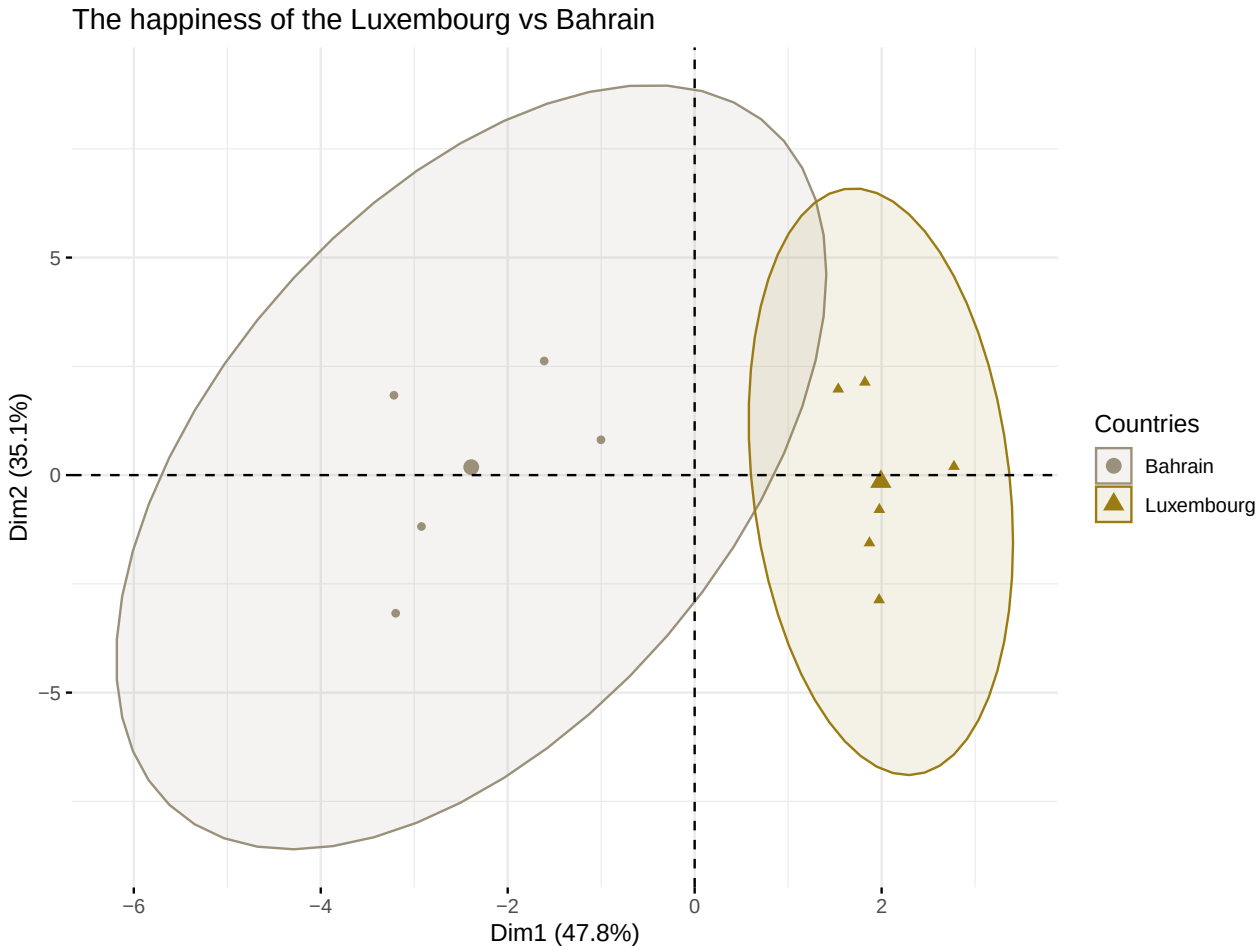
- **Costa Rica** the ellipse is extremely big showing a extremely large discrepancy.
- Dimension 1(PCA 1) the variance is both positive and negative on a more massive balanced scale based on the primary variable of happiness.
- Dimension 2(PCA 2) the variance of is both positive and negative on a massive balanced scale based on the secondary variable of happiness but slightly more on the positive side of dimension 2.
- **Australia** the ellipse is a little wide but thin so the discrepancy is a smaller for happiness.
- Dimension 1(PCA 1) the variance of happiness is mixed for both positive and negative for the primary variable.
- Dimension 2(PCA 2) the variance of the happiness is negative but not at a large scale for the secondary variable for happiness.



- **Austria** has a thin but extremely wide ellipse showing that the discrepancy is smaller.
- Dimension 1(PCA 1) is negative and their isn't much happiness in regard to the primary happiness variable.
- Dimension 2(PCA 2) is mixed for both negative and positive but mainly negative for the secondary primary variable.
- **Norway** has a much wider ellipse showing a discrepancy that is slightly smaller.
- Dimension 1(PCA 1) is mixed for both positive and negative where but mainly positive in regard to the primary happiness variable.
- Dimension 2(PCA 2) is mixed for both positive and negative where the secondary variable is mixed for both secondary variable of happiness.

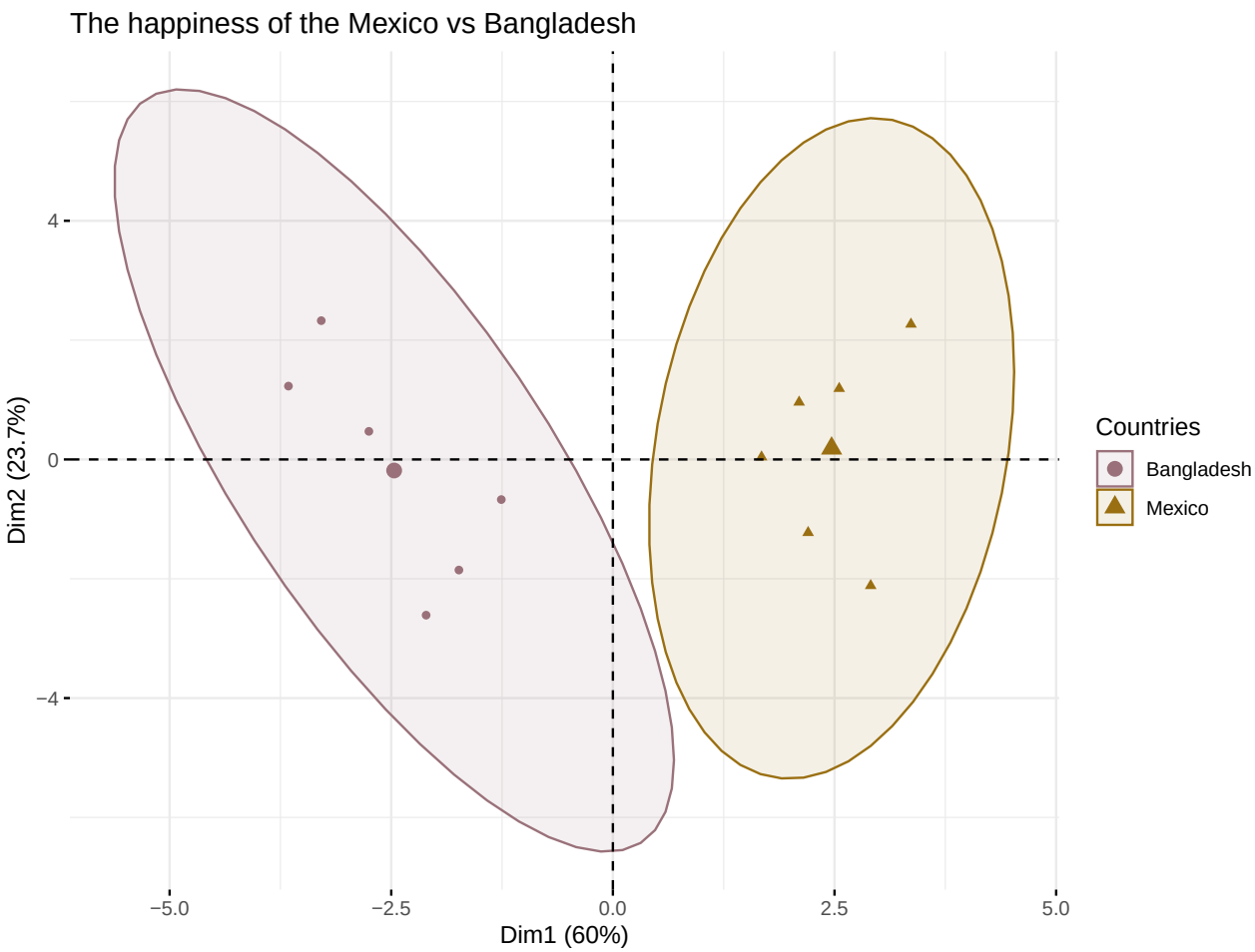


- **Israel** has a more thin ellipse and the discrepancy is smaller for happiness.
- Dimension 1(PCA 1) has a mixed balanced amount of happiness in regard the to the primary variable of happiness.
- Dimension 2(PCA 2) has positive average of happiness for the secondary variable for happiness.
- **Azerbaijan** has a more wide ellipse but not thin which means that the discrepancy is bigger for happiness.
- Dimension 1(PCA 1) has a mixed balanced amount of happiness for both negative and positive average over all for the primary variable for happiness.
- dimension 2(PCA 2) has a has a negative average of happiness for the secondary variable for happiness.



- **Luxembourg** has a small ellipse where their is a small amount of discrepancy.
- Dimension 1(PCA 1) has a mixed amount of happiness both negative and positive for the primary variable for happiness.
- Dimension 2(PCA 2) has a positive amount of happiness for the secondary variable.
- **Bahrain** has an extremely wide ellipse which accounts for the discrepancy for happiness.
- Dimension 1(PCA 1) has a balance negative and positive for the primary variable of happiness.

-Dimension 2(PCA 2) has a mainly negative happiness in regard to the secondary variable of happiness.



- **Mexico** has larger ellipse with a little bit discrepancy.
- Dimension 1(PCA 1) has a mixed amount of variance for both positive and negative in regard to the primary variable for happiness.
- Dimension 2(PCA 2) has a positive amount of happiness in regard to the secondary variable.
- **Bangladesh** has a large but also wide ellipse with a larger amount of discrepancy.
- Dimension 1(PCA 1) has a mixed amount of variance both positive and negative in regard to the primary variable for happiness.
- Dimension 2(PCA 2) has a mainly negative amount of happiness in regard to the secondary variable.

Based on PCA analysis of happiness indicators, countries show different patterns of happiness distribution among their populations:

Countries	Happiness varied/and how happy they are
United States	Happiness is more consistent it where some people experience a balanced amount of happiness
United Kingdom	Happiness is more varied where some people experience more happiness a lower rate compared others a low amount of happiness at a higher rate.
Finland	Happiness is much more varied where some people have more extreme amount of happiness and others have a much lower amount of happiness.

Countries	Happiness varied/and how happy they are
Afghanistan	Happiness is much more consistent and the happiness is much more balance were a few people is a little bit more happy compared to others who are a little more unhappy.
Denmark	Happiness is much more varied where the overall happiness is mixed and but lower over all.
Albania	Happiness is much better where happiness is more balanced and happiness is higher overall.
Iceland	Happiness is much more mixed at a balanced rate where the citizens are more unhappy overall.
Algeria	Happiness is more positive overall but has some areas where it is more mixed.
Argentina	Happiness is more mixed on average as a national whole is high.
Sweden	Happiness is mixed and the national average is lower as a whole.
Netherlands	Happiness is more mixed on average and the overall happiness is lower as a whole.
Armenia	Happiness is more mixed as a whole but overall happiness is higher.
Australia	Happiness is more mixed on average and the overall happiness is low but not on a large scale.
Costa Rica	Happiness is mixed on average and the scale of happiness in multiple factors is mixed but slight higher on the positive scale.
Austria	Happiness is mixed on average but mainly negative for the over all happiness.
Norway	Happiness is more mixed on average and the over all happiness is on a mixed scale with high amounts of happiness and low amount of happiness.
Israel	Happiness is more mixed on average with happiness being more positive.
Azerbaijan	Happiness is mixed on average with happiness being more negative.
Luxembourg	Happiness is more mixed on average with happiness being positive.
Bahrain	Happiness is more mixed on average with happiness being mainly negative.
Mexico	Happiness is more mixed on average with happiness being positive.
Bangladesh	Happiness is more mixed on average with happiness being negative.